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TransCelerate DDF Commercial Adaptation POC

This directory contains the Proof of Concept (POC) implementation for the commercial adaptation of the TransCelerate Digital Data Flow (DDF) repositories.

Overview

The POC demonstrates the technical feasibility and business value of adapting the DDF components for commercial use. It includes:

1. **SDR Core API** - A .NET 6 Web API that implements the core functionality of the Study Definition Repository
2. **StudyBuilder Adapter** - An adapter that integrates with upstream study design systems
3. **EDC Adapter** - An adapter that integrates with downstream Electronic Data Capture systems

Prerequisites

- [.NET 6 SDK](#)
- [Docker](#)
- [MongoDB](#) (or use the Docker container)

Project Structure

- /src - Source code for POC components
 - /SDR.Core.API - Core API implementation
 - /StudyBuilder.Adapter - Adapter for study design systems
 - /EDC.Adapter - Adapter for EDC systems
- /docs - Documentation and design artifacts
- /config - Configuration files
- /scripts - Setup and utility scripts
- /tests - Test cases and test data

Getting Started

Option 1: Running with Docker Compose

1. Make sure Docker is running
2. Open a terminal in the POC directory
3. Run the following command:

```
❏ash docker-compose up -d
```

This will start all components in Docker containers:

- MongoDB on port 27017
- SDR Core API on port 5000
- StudyBuilder Adapter
- EDC Adapter

Option 2: Running Locally

1. Make sure MongoDB is running on localhost:27017
2. Open a terminal in the POC directory
3. Run the following commands:

```
`❏`ash
```

Run SDR Core API

```
cd src/SDR.Core.API dotnet run
```

In a new terminal, run StudyBuilder Adapter

```
cd src/StudyBuilder.Adapter dotnet run
```

In a new terminal, run EDC Adapter

```
cd src/EDC.Adapter dotnet run `
```

Testing the POC

1. Access the SDR Core API Swagger UI at <http://localhost:5000/swagger>
2. Use the sample study definition in `/tests/sample-study.json` to create a new study
3. Verify the study is stored in MongoDB
4. Test the integration with the adapters

Implementation Details

SDR Core API

The SDR Core API provides endpoints for managing study definitions based on the USDM data model. It includes:

- CRUD operations for study definitions

- USDM data model implementation
- MongoDB integration for data storage
- Swagger documentation

StudyBuilder Adapter

The StudyBuilder Adapter integrates with upstream study design systems to retrieve study design data and transform it to the USDM format for storage in the SDR.

EDC Adapter

The EDC Adapter integrates with downstream Electronic Data Capture (EDC) systems to transform study definitions into EDC-specific formats.

Next Steps

After validating the POC, the next steps would be:

1. Expand the implementation to include more features
2. Integrate with additional systems
3. Implement a comprehensive BiomedicalConcept library
4. Develop a production-ready deployment strategy

Documentation

For more detailed information, see the following documents:

- [POC Strategy](#)
- [Architecture Document](#)